micro:bit

Introduction to Computer Science

Unit 5: Iteration

Microsoft

Lesson A
Understanding iteration

What we'll learn in Lesson A

- The value of iteration in programming
- Looping as a form of iteration

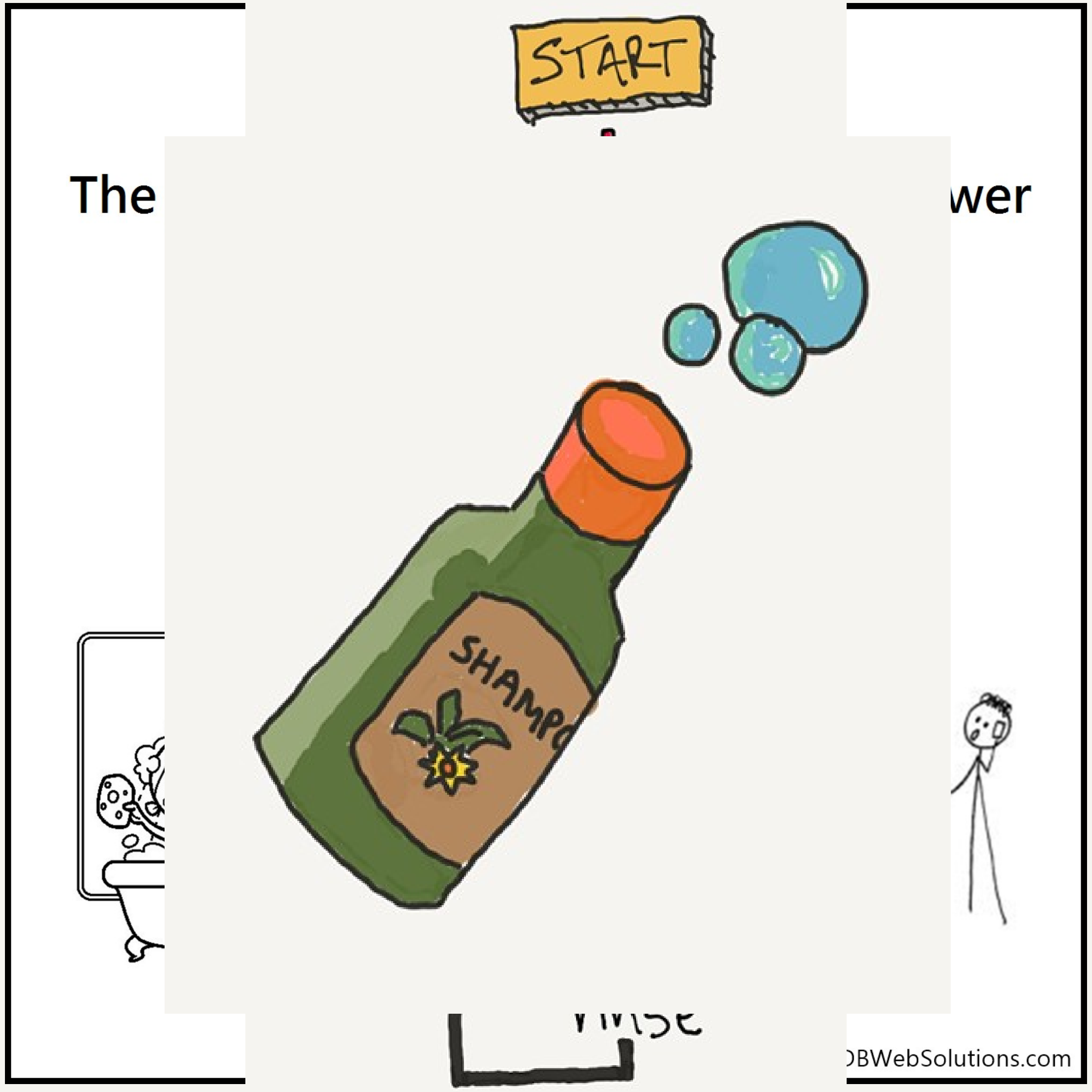
Iteration

Iteration is a fancy word for **Repetition**.

- You can “iterate” (or repeat) a process, or sequence of steps.
- In a computer program, iteration is done with the help of “Loops”.
- A Loop allows a computer to repeat lines of code.

1. What algorithm could you write for shampooing your hair?

2. What loops would you use?



Beware of infinite loops!

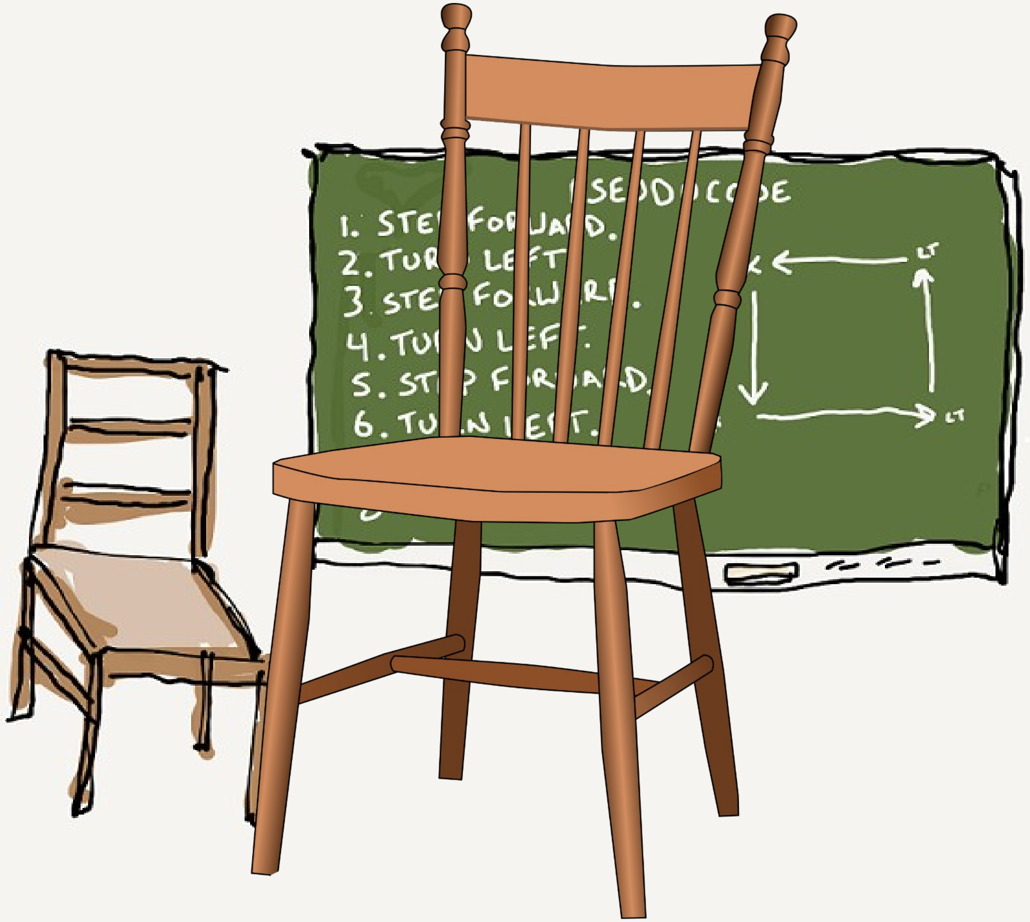
Activity: Walk a square

Write pseudocode instructions to tell me how to walk around a chair.

- How many lines of code is this?
- Can we be more efficient using a loop?

Repeat (4 times):

- Step forward
- Turn left



Microsoft

What we learned in Lesson A

- The value of iteration in programming
- Looping as a form of iteration

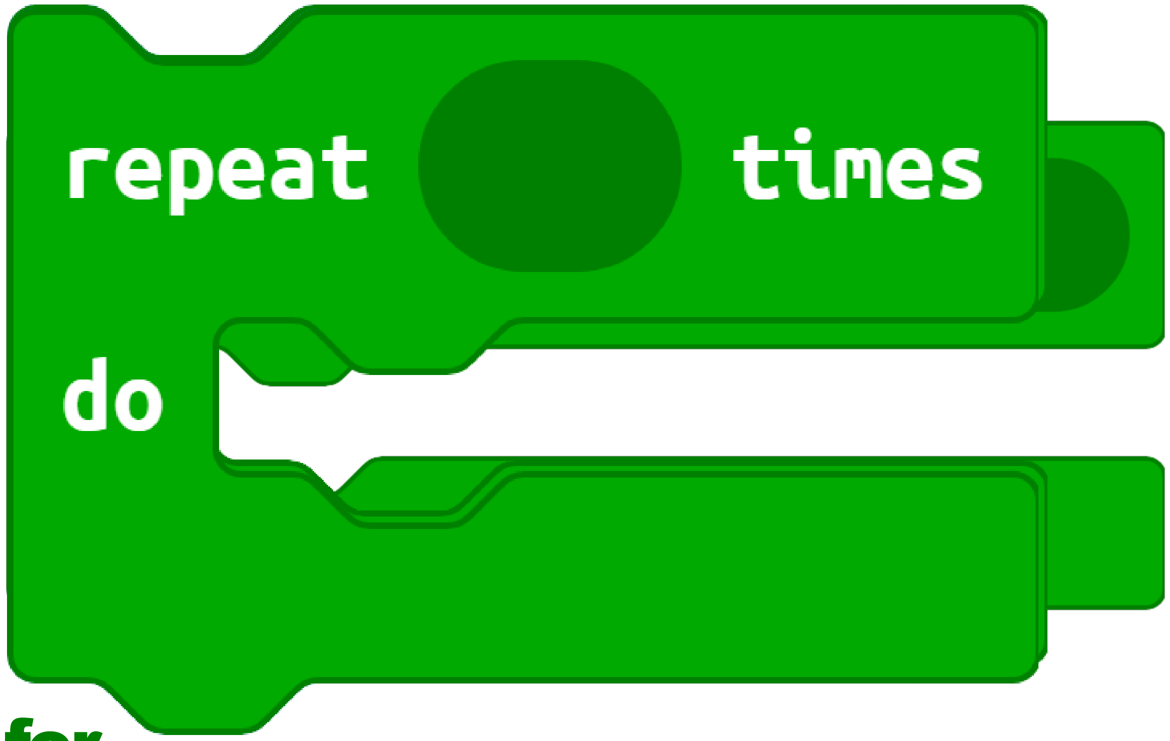
Next time: Lesson B

- Code with loops
- Assess our learning with a quiz

Lesson B
Coding with loops

What we'll learn in Lesson B

Use the different loop blocks in coding



for

repeat Repeat code a given number of times

while Repeat code while a condition is true.

Using an index

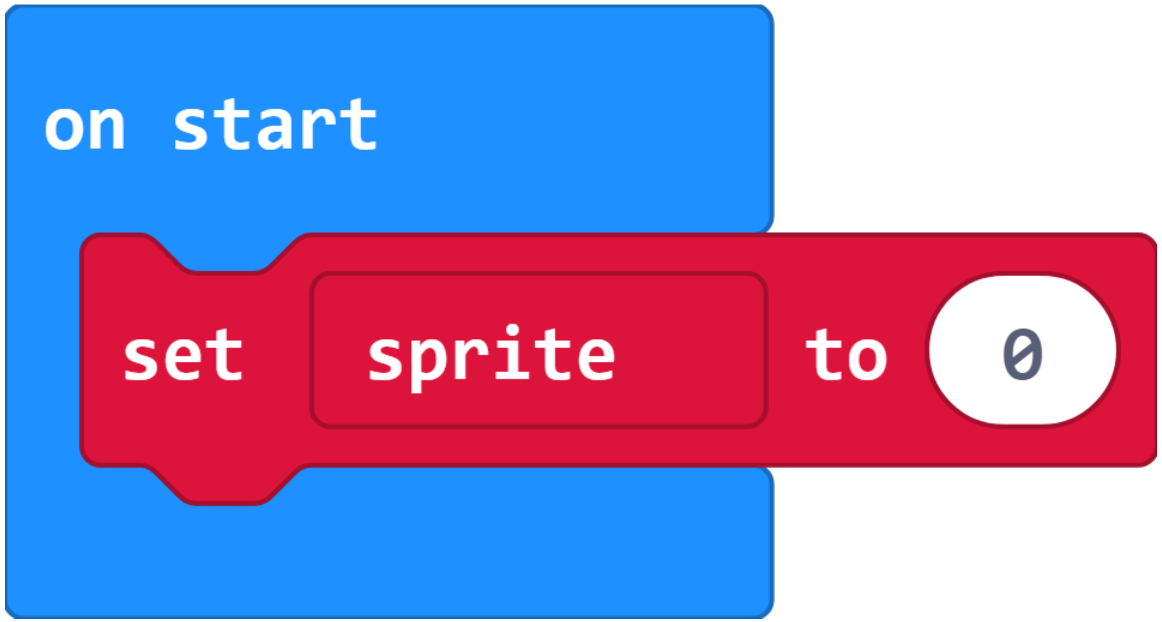
Use the Repeat loop: Steps 1 & 2

In this program, we'll use the **Repeat loop** to make a Sprite walk a square on the micro:bit.

1. Create a new project and call it something like "**Sprite walking square**".

2. From the Variables Toolbox drawer, make a variable named "**sprite**".

Drag out a **set variable** block into the **on start** block.



Step 3

a. Click on the Advanced Toolbox drawer to expand the Toolbox.

b. Click on the Game Toolbox drawer and drag out a **Create Sprite block**.

c. Drop it into the **Set variable** block replacing 0.

d. Set the starting coordinates at **0, 0** (top left corner).

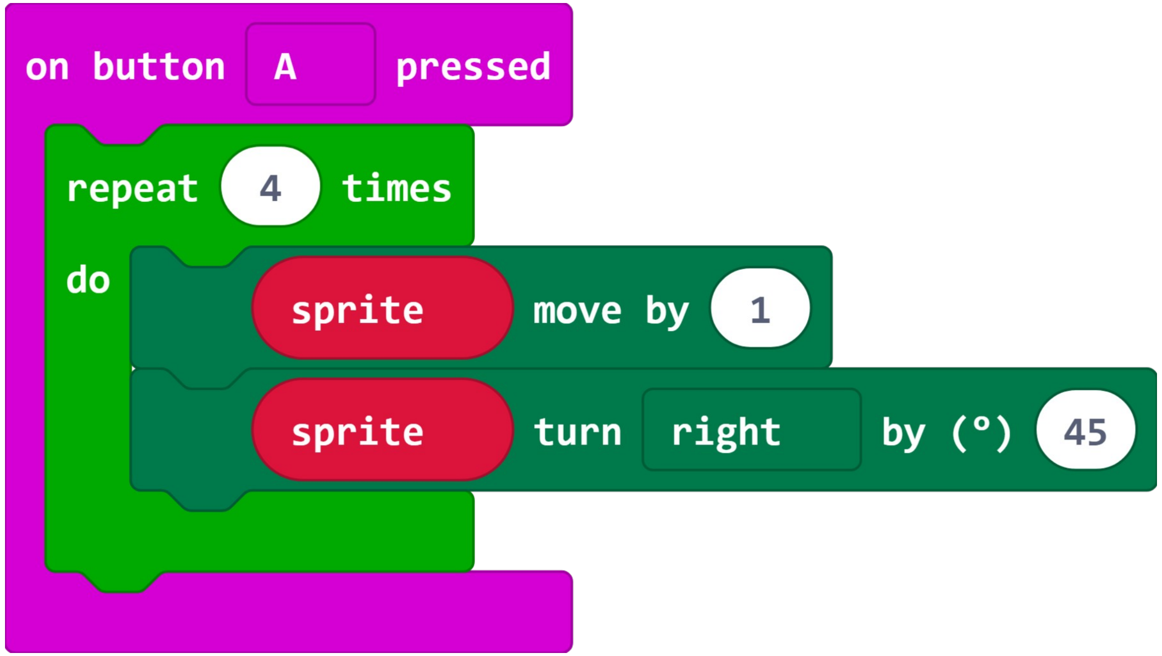
Steps 4 & 5

4. From the Input Toolbox drawer, drag out an **on button pressed** block.

   From the Loops Toolbox drawer drag out a **Repeat** loop.

4. From the Game Toolbox drawer drag out two blocks: **Sprite Move** and **Sprite Turn**.

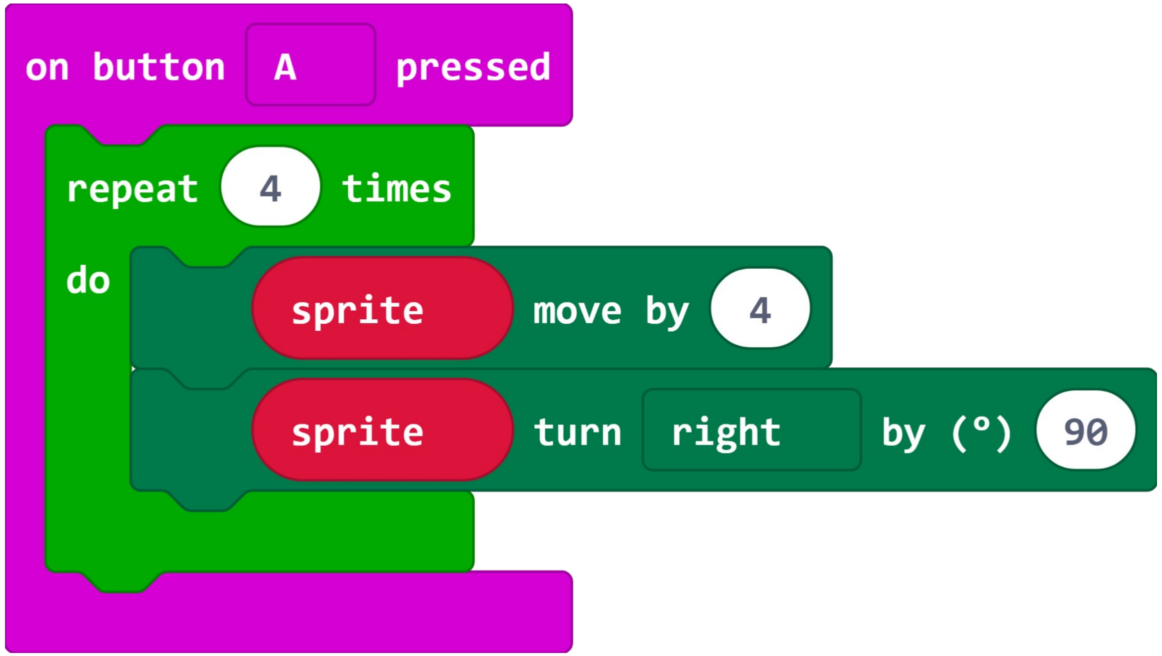
   Drop these blocks inside the **Repeat** loop.



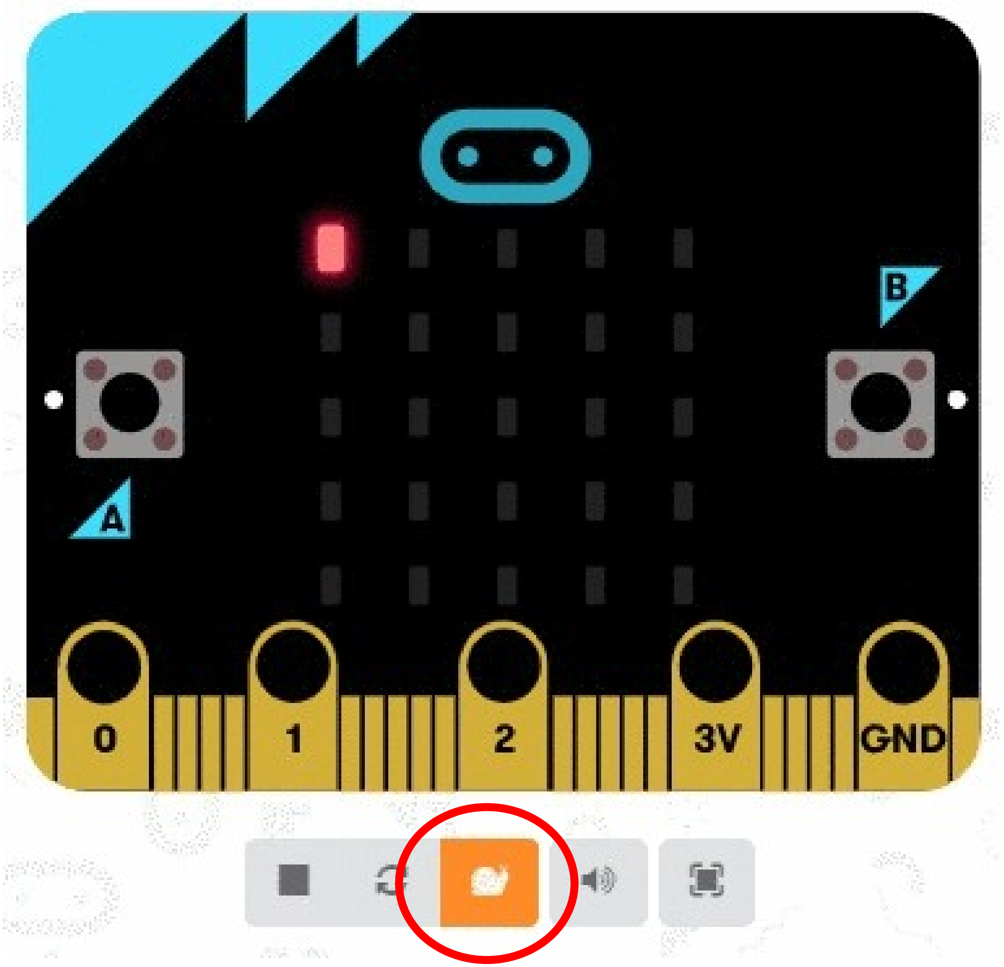
Steps 6 & 7

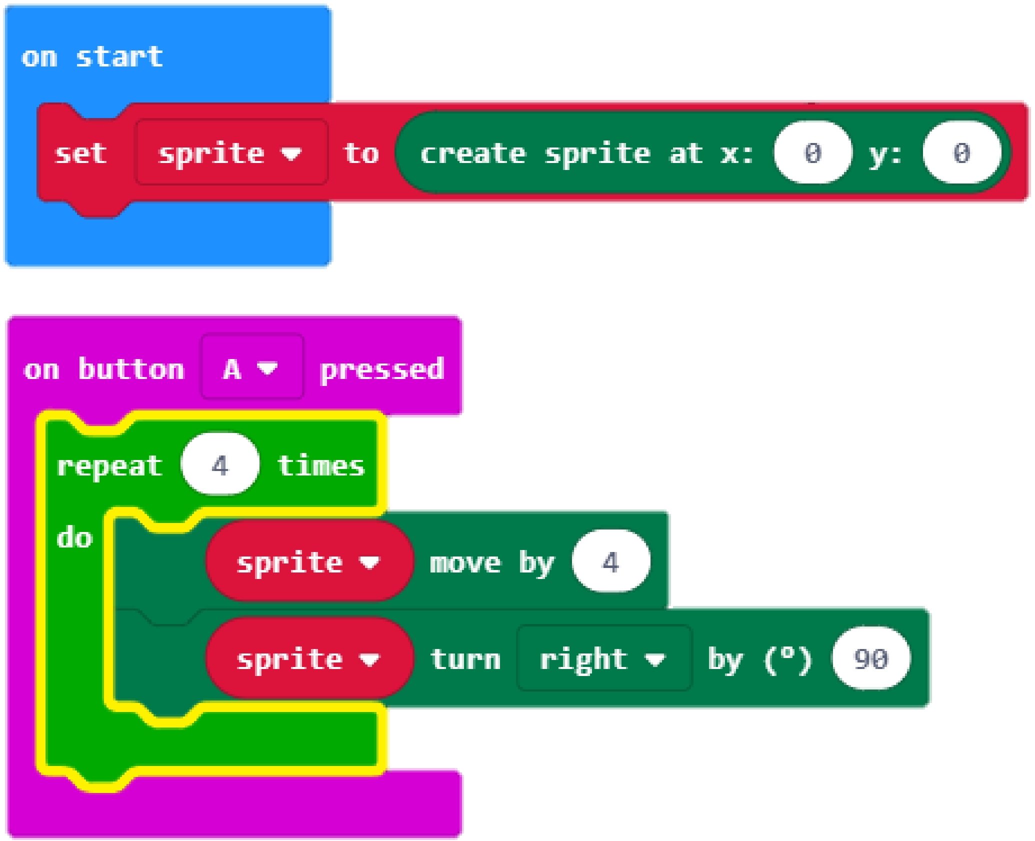
6. Change the value in the **sprite move** block four spaces.

6. Change the value in the **sprite turn** block to 90 degrees.



The Slo-Mo button





Select the Slo-Mo button in the Simulator to see your code running

Use the For loop

The **For loop** is useful when you have a variable in your loop that you want to increment each time through.

In this program, we'll use the For loop to make an LED light move across the screen from left to right and top to bottom.

Write the pseudocode instructions for this program first.

Pseudocode for the first row:

- `Turn on  (0, 0)`
- `Turn off (0, 0)`
- `Turn on  (1, 0)`
- `Turn off (1, 0)`
- `Turn on  (2, 0)`
- `Turn off (2, 0)`
- `Turn on  (3, 0)`
- `Turn off (3, 0)`
- `Turn on  (4, 0)`
- `Turn off (4, 0)`

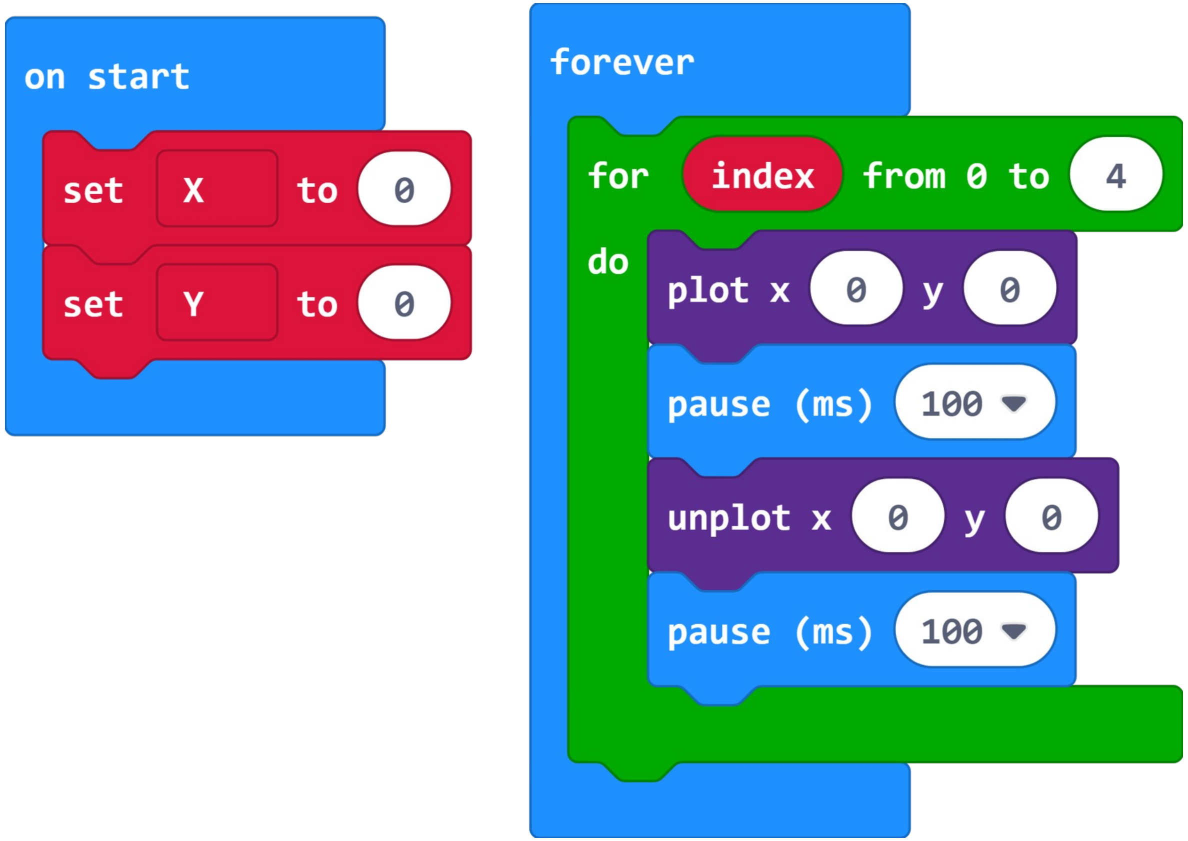
1. **What is the only variable that is changing in this pseudocode?**
2. **How would you write the pseudocode for the second, third, and fourth rows?**

For loop coding activity

Now try writing a program in MakeCode

Hints:

- Make **2 variables** to hold the X and Y coordinate values.
- Use those variables as the **index** of your For loops.
- Use the **Plot** and **Unplot** LED blocks (use **Pause** blocks in between).
- You will need to create **nested For loops** for both the X and Y coordinates.

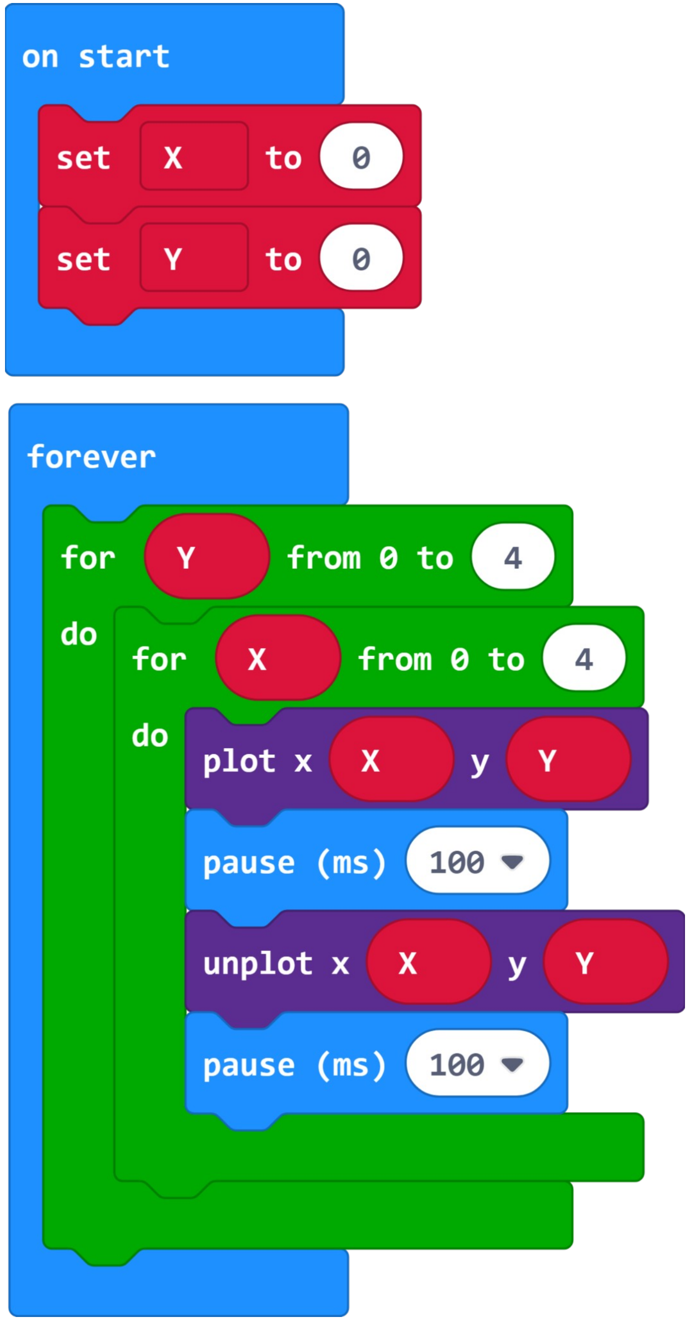


Your final program

Your final program may look like the example on the right.

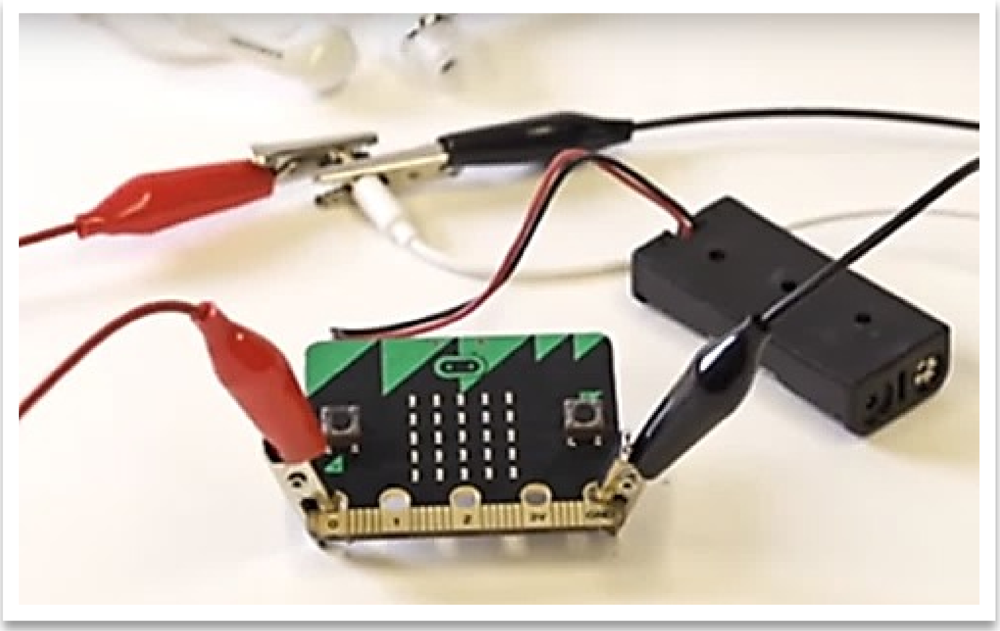
Optional Mods:

- Experiment with changing the parameters to see how these changes affect the program.
- What happens if you switch the positions of the nested loops, so the outer loop loops through the 'X' index values and the inner loop loops through the 'Y' index values?
- What happens if you remove the 'unplot' block and the 'pause' block below it?



Use the While loop

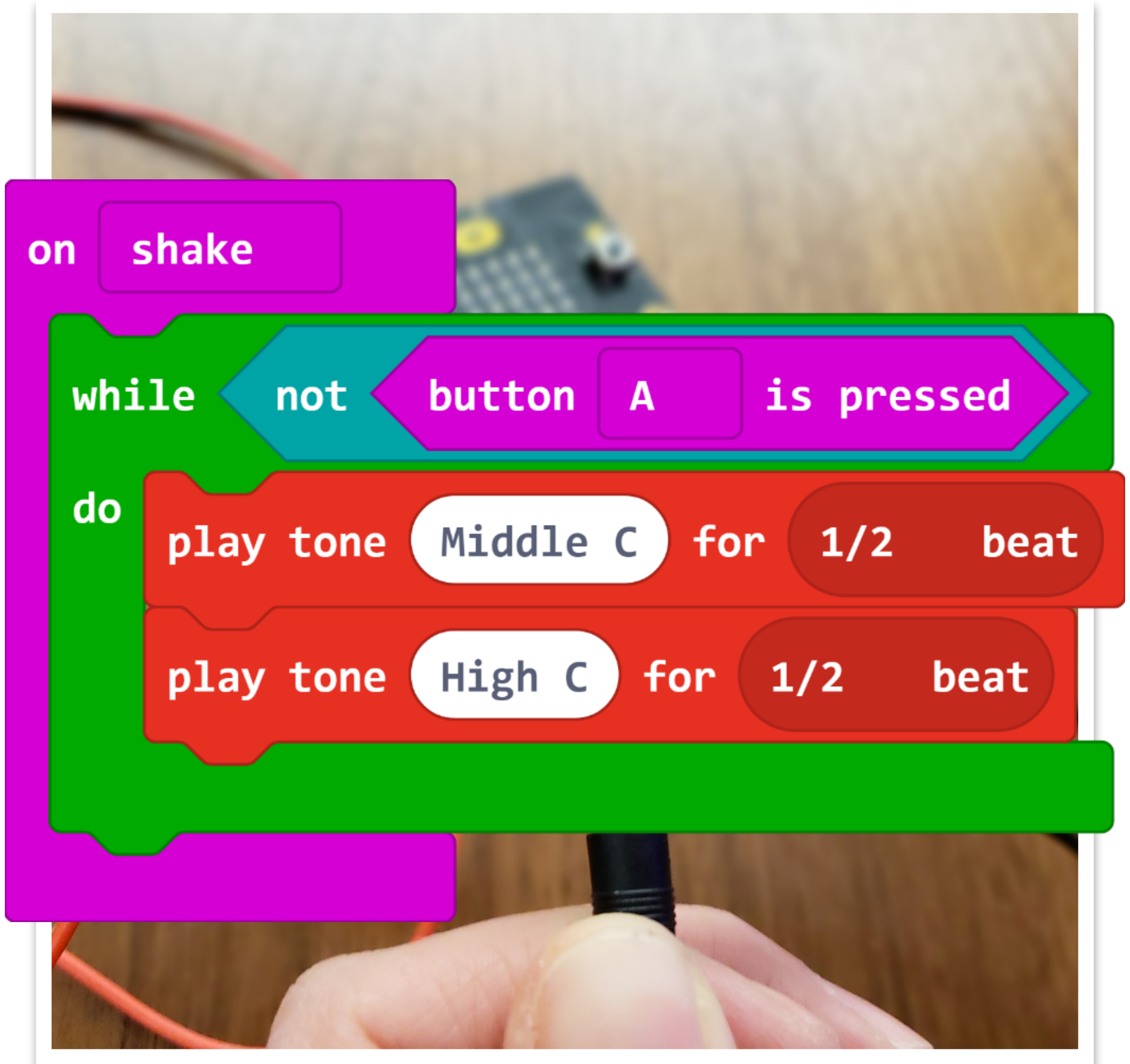
- The While loop is useful when you want your program to loop until a certain event happens or a different condition is met.
- In this program, we'll use the While loop to create a micro:bit alarm. The alarm will sound if someone shakes the micro:bit, and it will continue to sound until someone presses a button.
- You will need to connect headphones or speakers to your micro:bit in order to play sounds.



While loop final program

Your final program may look like the example on the right.

Use Croc Clips to connect P0 and GND pins on the micro:bit to the headphone jack.



Time for a quiz!

What we learned in Lesson B

- Used three different types of loop blocks in coding
- How much we've already learned

Next time: Lesson C

- Get loopy!

Lesson C
Get loopy!

What we'll learn in Lesson C

- Create a program that uses loops, variables, and parameters
- Design and build an object that uses the micro:bit program with sound, display, and motion

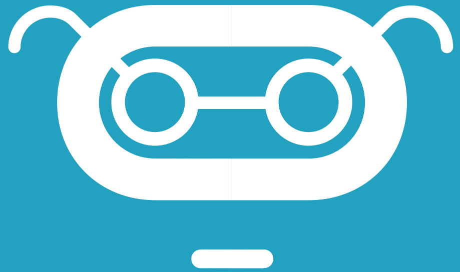
Activity: Get loopy!

There are many different ways to use the three types of loop blocks.

- How will you use loops to create something useful, entertaining, or interesting?
- What might you make?

Some project ideas

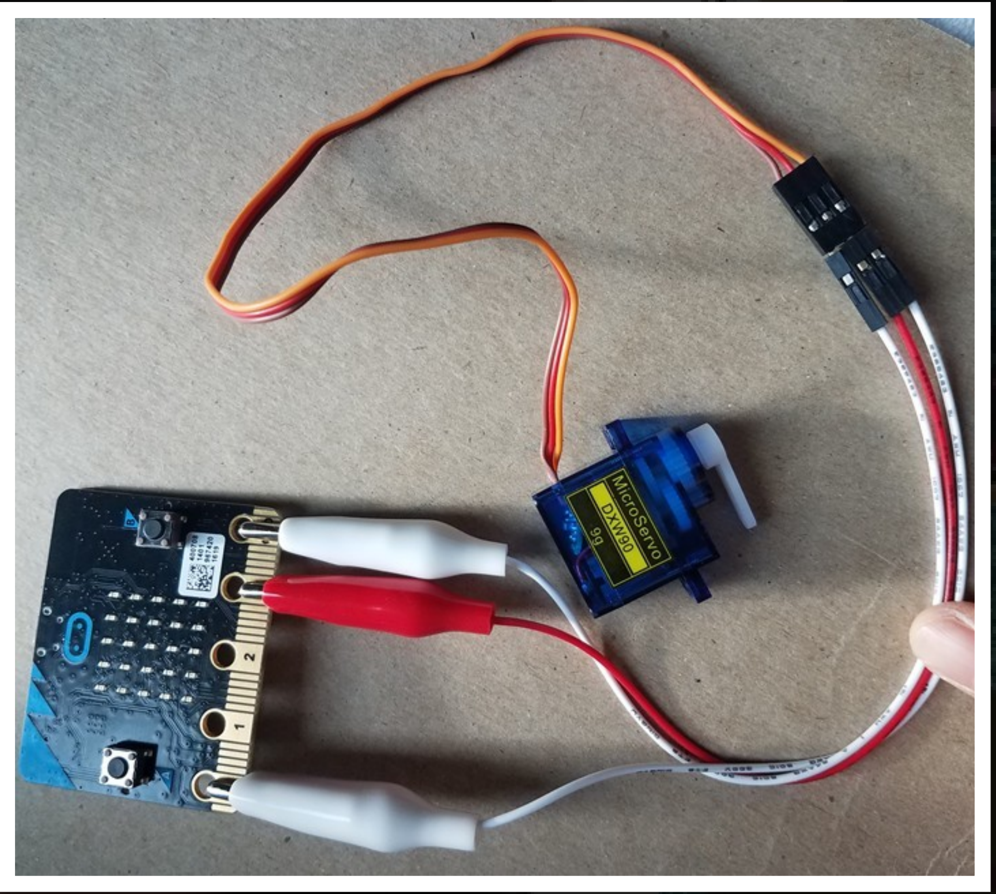
- Create an animated GIF (looping image that changes) and add music that matches.
- Create animation that repeats for one of the melodies included in MakeCode (like Happy Birthday).
- Create different animations that run when different buttons are pressed.
- Create an alarm that includes sound and images. What will set the alarm off? What will make the alarm stop sounding?
- Use servo motors to create a creature that dances and changes its expression while a song plays.



Project example: A musical birthday card

- Press A to make a wish
- Press B to hear a surprise song

Project example: micro:bit hat man



This micro:bit is connected to a servo motor with Crocodile clips.

This project uses the micro:bit light sensor to display a happy face when it's sunny, and a frowning face when it's dark.

Project expectations

Use a design thinking approach and ensure the project meets these specifications:

- Uses at least three different loops in a meaningful way
- Uses unique variable names that clearly describe the variable values
- Uses sound, display, and motion in a way that's integral to the program
- The program compiles and runs as intended and includes meaningful comments in code
- Provide the written Reflection Diary entry

Reflection Diary

- Explain how you decided on your particular “loopy” idea. What brainstorming ideas did you come up with?
- What type of loop did you use: For, While, or Repeat?
- What was something surprising to you about the process of creating this program?
- Describe a difficult point in the process of designing this program and explain how you resolved it.
- What feedback did your beta testers give you? How did that help you improve your loop demo?
- Publish your MakeCode program and include the link.

What we learned in Lesson C

- Created a program that uses loops, variables, and parameters
- Designed and built an object that uses the micro:bit program with sound, display, and motion